

ONNX GenAI Working Group

Year in Review (June 2025 – May 2026)

29

MEETINGS COMPLETED

Bi-Weekly

CADENCE STATUS

Multi-Org

ACTIVE COLLABORATION

Chairs: Yamini Nimmagadda, Javier Martinez, G. Ramalingam

OVERVIEW

- Goal: Enable recent GenAI models
- Technical focus: overlaps other SIGs
 - Operator SIG
 - Converter SIG
- + Other new areas

OPERATORS STANDARDIZED

- TensorScatter
 - To represent in-place updates to KV Cache
- LinearAttention
 - New attention op used by Mamba, Qwen 3.5, & Nemotron
- FlexAttention
 - As a generic, higher-order op, to express variations of Attention
 - In experimental domain

ONGOING

- Quantization
 - Optimized storage format for ternary-valued tensors
- Grouped MatMul
 - Expert selection for Mixture Of Experts (MoE)
- Paged Attention
 - For efficient memory management for large models

Streamlining LLM Exporters

✓ Optimum ONNX:

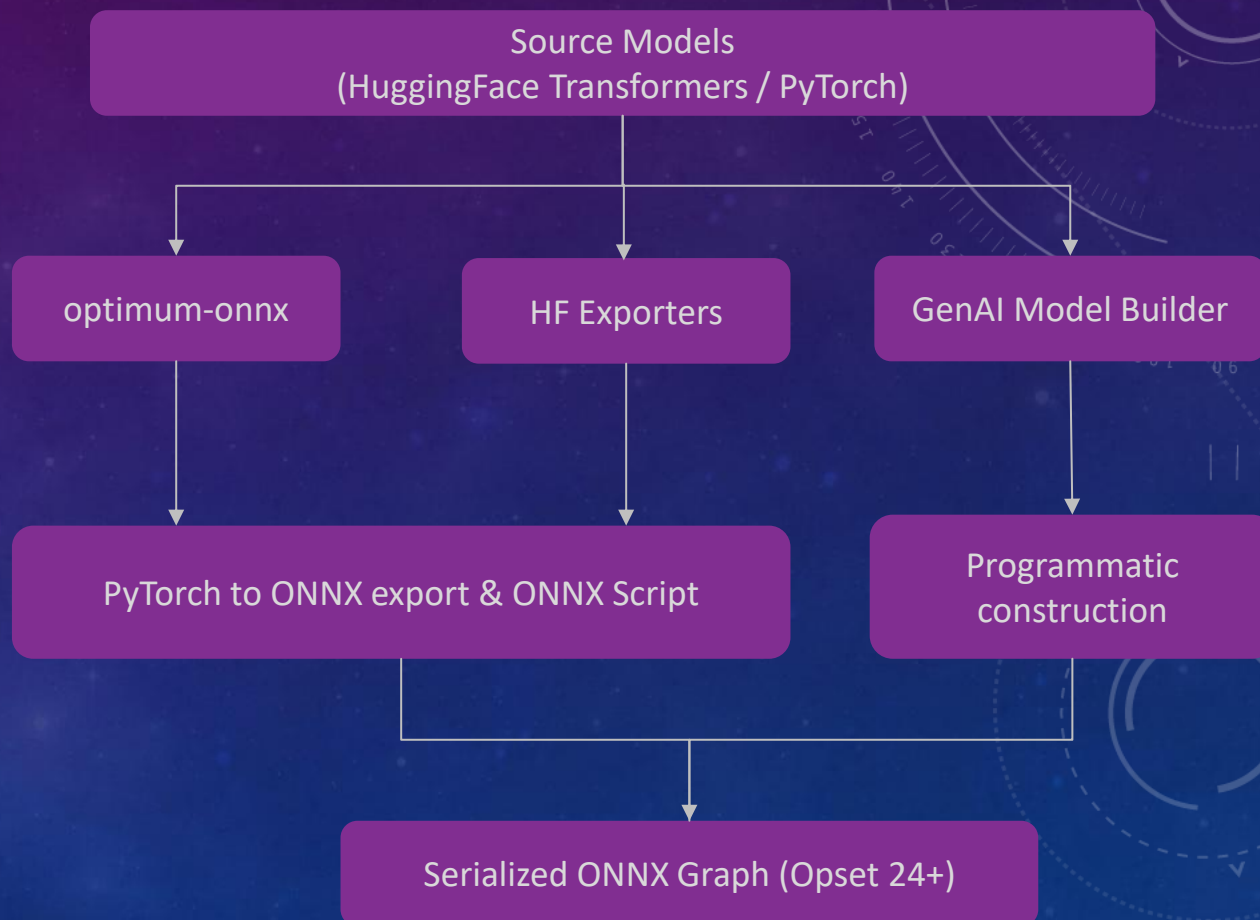
- Updated ONNX exporter version and raised PR (Xavier) to migrate to torchdynamo based export
- Executed gap analysis targeting missing ORT classes
- Discussed observer-based export proposal from Xavier

✓ HF Exporters:

- Discussed with HuggingFace on using direct export pipeline from Hugging Face Transformers utilizing TorchDynamo
- Conducted analysis to test the maturity of the ONNX export and identified gaps for VLM model classes with this approach
- Tracking HF exporters as a key export path for ONNX

✓ ONNX GenAI Model Builder:

- Discussed GenAI model builder as one of the approaches to generate ONNX models using programmatic model construction



ENABLING BACKEND-SPECIFIC OPTIMIZATIONS

- RFC #7301 — Equivalent subgraphs and multi-target model representations.
- Packager pattern — Keep the standard graph; attach backend-optimized variants alongside.
- Weight reuse — Allow weight sharing across variants for smaller disk size
- Compatibility strings — Lightweight tags to validate pre-compiled blobs at load time.

ENABLING AOT COMPILATION

- Backend Context: custom ops that carry (compiled) binary data
- Compatibility strings — Lightweight tags to validate pre-compiled blobs at load time.

GEN-AI INTERFACES

- Standardized GenAI API
 - A unified C++ API for shipping GenAI models everywhere.
- Goal — Framework- and platform-independent C++ surface for GenAI deployment.
- Target — Client and edge PCs with heterogeneous NPU, GPU, and CPU hardware.
- Proof points — Working text-to-text and text-to-image [pipeline interfaces](#).
- Outcome— Prompted discussions and interface design efforts within ONNX Runtime GenAI and WinML

Looking Ahead

Thank you.

For your contributions to streamlining the GenAI flow.

GenAI WG Continues Monthly

We will keep the GenAI WG running on a monthly cadence with **focused discussions and an agenda shared ahead of time.**

Targeted Focus Areas

New GenAI operations, ultra-low precision support and storage, ONNX exporter workflows, backend context, and related items.